

Denoising Diffusion Probabilistic Models

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Introduction

- Denoising Diffusion Probabilistic Models (DDPMs) are a class of deep generative models—machine learning systems trained to synthesize new data that closely resembles their training distribution.
- The method consists of two phases: (1) a **forward diffusion process** that progressively adds Gaussian noise to data over many timesteps until the signal becomes pure noise, and (2) a **reverse process**, learned by a neural network, that iteratively denoises random noise to generate new samples.

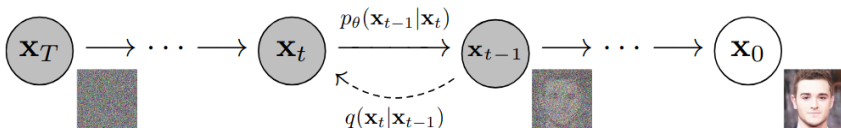


Figure: forward diffusion process and reverse process

Forward Diffusion

- Given an image $\mathbf{x}_0 \in \mathbb{R}^d$, the *forward process* iteratively adds noise to create a sequence $\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_T$ via the recurrence for the measurable function f :

$$\mathbf{x}_t = \sqrt{1 - \beta_t} \mathbf{x}_{t-1} + \sqrt{\beta_t} \epsilon_t = f(\mathbf{x}_{t-1}, \epsilon_t) \quad (1)$$

Here, each $\epsilon_t \stackrel{i.i.d}{\sim} \mathcal{N}(\mathbf{0}, \mathbf{I}_d)$, and $\beta_t \in (0, 1)$ is a pre-specified scalar called the variance of the added noise in the update at step t . The collection $\{\beta_t\}_{t=1}^T$ is increasing.

- Given a data distribution $\mathbf{x}_0 \in \mathbb{R}^d \sim q(\mathbf{x}_0)$, the forward Markov process generates a sequence of random variables $\mathbf{x}_1, \mathbf{x}_2 \dots \mathbf{x}_T$ with transition kernel $q(\mathbf{x}_t | \mathbf{x}_{t-1})$. The joint distribution of $\mathbf{x}_1, \mathbf{x}_2 \dots \mathbf{x}_T$ conditioned on $\mathbf{x}_0$, denoted as $q(\mathbf{x}_1, \dots, \mathbf{x}_T | \mathbf{x}_0)$ is

$$q(\mathbf{x}_{1:T} | \mathbf{x}_0) := q(\mathbf{x}_1, \dots, \mathbf{x}_T | \mathbf{x}_0) = \prod_{t=1}^T q(\mathbf{x}_t | \mathbf{x}_{t-1}) \quad (2)$$

where $q(\mathbf{x}_t | \mathbf{x}_{t-1}) = \mathcal{N}(\mathbf{x}_t; \sqrt{1 - \beta_t} \mathbf{x}_{t-1}, \beta_t \mathbf{I})$

Forward Diffusion

- Analytical form of $q(\mathbf{x}_t \mid \mathbf{x}_0)$ for all $t \in \{0, 1, \dots, T\}$. Specifically, denoting $\alpha_t := 1 - \beta_t$ and $\bar{\alpha}_t := \prod_{s=0}^t \alpha_s$, we have

$$q(\mathbf{x}_t \mid \mathbf{x}_0) = \mathcal{N}(\mathbf{x}_t; \sqrt{\bar{\alpha}_t} \mathbf{x}_0, (1 - \bar{\alpha}_t) \mathbf{I}) \quad (3)$$

- Given $\mathbf{x}_0 \sim q(\mathbf{x}_0)$, we can easily obtain a sample of $\mathbf{x}_t$ by sampling a Gaussian vector $\epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ and applying the transformation yields

$$\mathbf{x}_t \stackrel{(d)}{=} \sqrt{\bar{\alpha}_t} \mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon \quad (4)$$

When $\bar{\alpha}_T \approx 0$, $\mathbf{x}_T$ is almost Gaussian in distribution, so we have

$$q(\mathbf{x}_T) \approx p_{\text{prior}}(\mathbf{x}_T) = \mathcal{N}(\mathbf{x}_T; \mathbf{0}, \mathbf{I}),$$

with $q(\mathbf{x}_T) = p_{\text{prior}}(\mathbf{x}_T)$ in the limit as $T \rightarrow \infty$. Here, p_{prior} denotes the *prior distribution* over the final latent variable $\mathbf{x}_T$, which is conventionally chosen to be a standard isotropic Gaussian.

Reverse Process

- **Goal:** Sample from the data distribution $q(\mathbf{x}_0)$ by reversing a noising process.
- **Problem:** The true reverse transition $q(\mathbf{x}_{t-1} | \mathbf{x}_t)$ requires $q(\mathbf{x}_0)$ and involves intractable integrals:

$$q(\mathbf{x}_{t-1} | \mathbf{x}_t) = \frac{q(\mathbf{x}_t | \mathbf{x}_{t-1}) \int q(\mathbf{x}_{t-1} | \mathbf{x}_0) q(\mathbf{x}_0) d\mathbf{x}_0}{\int q(\mathbf{x}_t | \mathbf{x}_0) q(\mathbf{x}_0) d\mathbf{x}_0}$$

→ We don't know $q(\mathbf{x}_0)$ (p_{complex}), and the integrals are impossible to compute.

- **Solution:** Approximate the reverse process with a neural network:

$$p_{\theta}(\mathbf{x}_{t-1} | \mathbf{x}_t) = \mathcal{N}(\mu_{\theta}(\mathbf{x}_t, t), \Sigma_{\theta}(\mathbf{x}_t, t))$$

The network learns to denoise step-by-step using only $\mathbf{x}_t$ and t .

- **Reverse process** is a Markov chain:

$$p_{\theta}(\mathbf{x}_{0:T}) = p(\mathbf{x}_T) \prod_{t=1}^T p_{\theta}(\mathbf{x}_{t-1} | \mathbf{x}_t),$$

with $p(\mathbf{x}_T) = \mathcal{N}(\mathbf{0}, \mathbf{I})$.

Methodology

Definition (KL divergence)

Let $(\mathcal{X}, \mathcal{F})$ be a measurable space and $\mathbb{P}$ and $\mathbb{Q}$ are its probability measures. If both $\mathbb{P}$ and $\mathbb{Q}$ are both absolutely continuous with respect to the Lebesgue measure μ , then there exist densities p and q respectively such that

$$D_{\text{KL}}(\mathbb{Q} \parallel \mathbb{P}) = \int_{\mathbf{x} \in \mathcal{X}} q(\mathbf{x}) \log \frac{q(\mathbf{x})}{p(\mathbf{x})} \mu(d\mathbf{x}), \quad \text{when } \mathbb{Q} \ll \mathbb{P}$$

In Generative Modeling, we want to learn a model $p_{\theta}(\mathbf{x}_0)$ that approximates the real data complex distribution $q(\mathbf{x}_0)$, so that we can generate new samples of data. One can show that

$$\arg \min_{\theta} \mathcal{D}_{\text{KL}}(q(\mathbf{x}_0) \parallel p_{\theta}(\mathbf{x}_0)) = \arg \max_{\theta} \mathbb{E}_{\mathbf{x}_0 \sim q(\mathbf{x}_0)} [\log p_{\theta}(\mathbf{x}_0)]$$

This shows minizing the KL-divergence is equivalent to performing the MLE:

$$\theta^* = \arg \max_{\theta} \mathbb{E}_{q(\mathbf{x}_0)} [\log p_{\theta}(\mathbf{x}_0)]$$

- However, the marginal likelihood

$$p_{\theta}(\mathbf{x}_0) = \int_{\mathbf{x}_{1:T}} p_{\theta}(\mathbf{x}_{0:T}) d\mathbf{x}_{1:T}.$$

is intractable: the latent trajectory $\mathbf{x}_{1:T}$ is high-dimensional, and the neural reverse process prevents closed-form integration. Thus, $\log p_{\theta}(\mathbf{x}_0)$ cannot be computed directly, but we can optimize a variational lower bound for $\mathbb{E}_{q(\mathbf{x}_0)}[\log p_{\theta}(\mathbf{x}_0)]$.

Theorem (Evidence lower bound (ELBO) on log likelihood of DDPM)

$$\begin{aligned} & \mathbb{E}_{q(\mathbf{x}_0)} [-\log p_{\theta}(\mathbf{x}_0)] \\ & \leq \mathbb{E}_{q(\mathbf{x}_0)q(\mathbf{x}_{1:T}|\mathbf{x}_0)} \left[\mathcal{D}_{KL}(q(\mathbf{x}_T | \mathbf{x}_0) \| p(\mathbf{x}_T)) + \sum_{t=2}^T \mathcal{D}_{KL}(q(\mathbf{x}_{t-1} | \mathbf{x}_t, \mathbf{x}_0) \| p_{\theta}(\mathbf{x}_{t-1} | \mathbf{x}_t)) \right. \\ & \quad \left. - \log p_{\theta}(\mathbf{x}_0 | \mathbf{x}_1) \right] \end{aligned}$$

Theorem (Evidence lower bound (ELBO) on log likelihood of DDPM)

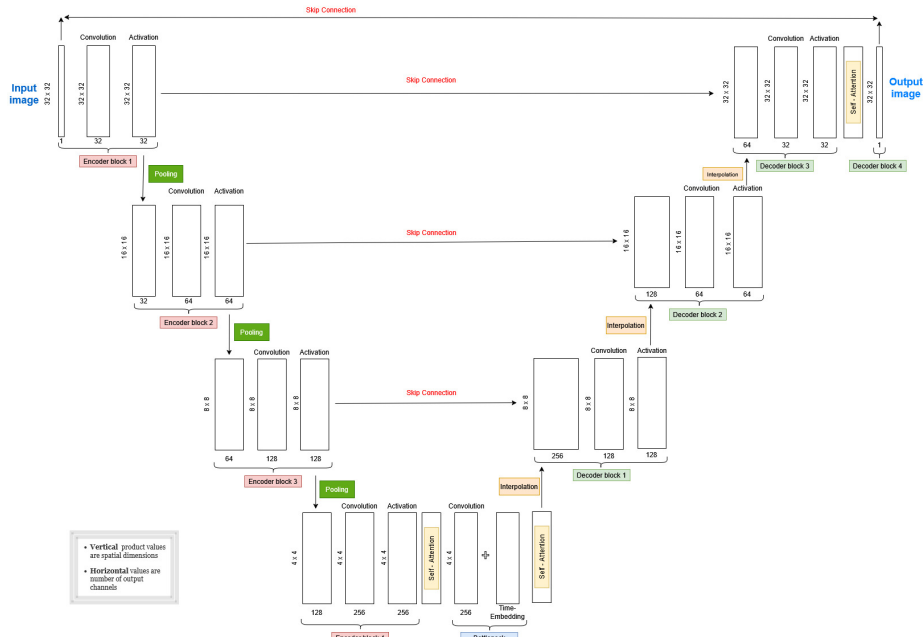
$$\begin{aligned} & \mathbb{E}_{q(\mathbf{x}_0)} [-\log p_\theta(\mathbf{x}_0)] \\ & \leq \mathbb{E}_{q(\mathbf{x}_0)q(\mathbf{x}_{1:T}|\mathbf{x}_0)} \left[\mathcal{D}_{KL}(q(\mathbf{x}_T | \mathbf{x}_0) \| p(\mathbf{x}_T)) + \sum_{t=2}^T \mathcal{D}_{KL}(q(\mathbf{x}_{t-1} | \mathbf{x}_t, \mathbf{x}_0) \| p_\theta(\mathbf{x}_{t-1} | \mathbf{x}_t)) \right. \\ & \quad \left. - \log p_\theta(\mathbf{x}_0 | \mathbf{x}_1) \right] \end{aligned}$$

- To simplify the training loss, the model is trained with the simple loss function to minimize:

$$L_{\text{simple}}(\theta) := \mathbb{E}_{t \sim \mathcal{U}[1, T], \mathbf{x}_0 \sim q(\mathbf{x}_0), \epsilon \sim \mathcal{N}(0, \mathbf{I})} \left[\left\| \epsilon - \epsilon_\theta(\sqrt{\bar{\alpha}_t} \mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon, t) \right\|^2 \right]$$

Results

- We fix the forward process variances $\{\beta_t\}_{t=1}^T$ to constants that increase linearly from $\beta_1 = 10^{-4}$ to $\beta_T = 0.02$ across all models in our experiments.
- The reverse (denoising) process is modeled using a U-Net architecture. We train all models with the Adam optimizer using a constant learning rate of 10^{-3} .



MNIST

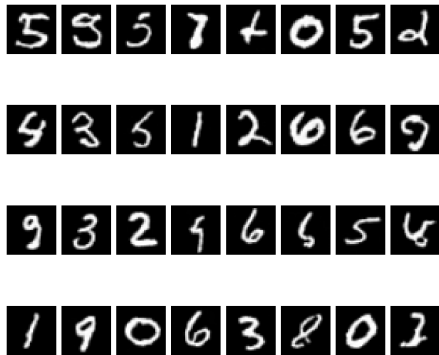


Figure: Diffusion_model_2 generated sample with $T = 500$.

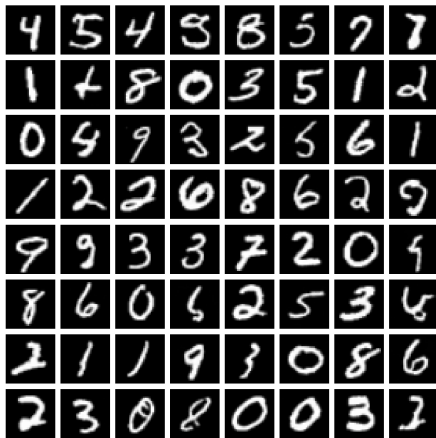


Figure: Diffusion_model_2 comparison with real MNIST samples

CIFAR-10 Reverse Process for Image Generation

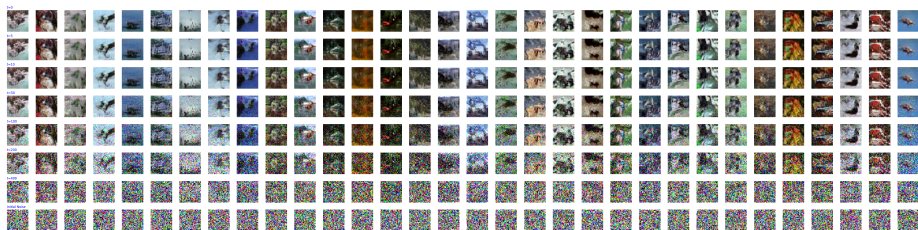


Figure: CIFAR-10_diffusion_model_3 Progressive Generation

CIFAR-10

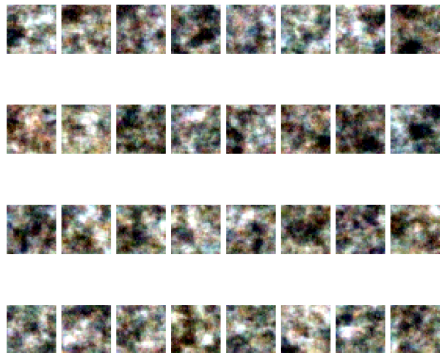


Figure: CIFAR-10_diffusion_model_1
 generated samples with $T = 500$.



Figure: Real vs Generated samples
 comparison from Figure 7

CIFAR-10

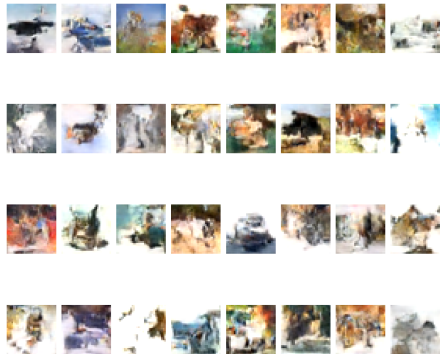


Figure: CIFAR-10_diffusion_model_2 generated samples with $T = 1000$.

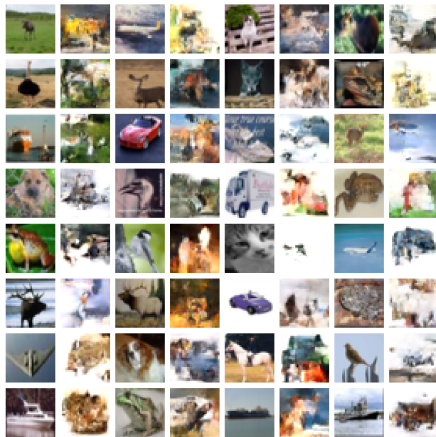


Figure: Real vs Generated samples comparison from Figure 9

CIFAR-10

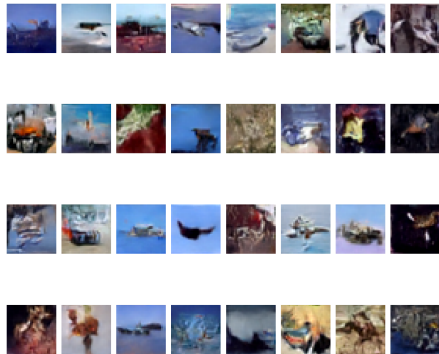


Figure: CIFAR-10_diffusion_model_3
 generated samples with $T = 1000$.

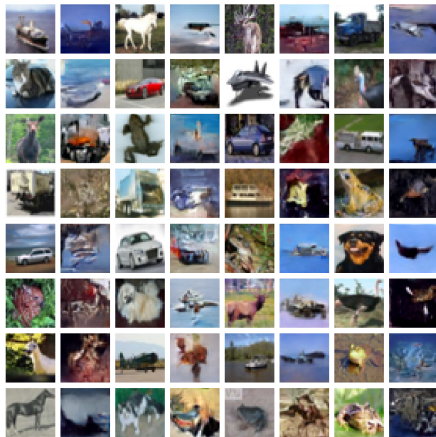


Figure: Real vs Generated samples
 comparison from Figure 11

Thank you for listening

References

Jonathan Ho, Ajay Jain, and Pieter Abbeel. (2020). Denoising Diffusion Probabilistic Models. arXiv preprint arXiv:2006.11239

Jascha Sohl-Dickstein, Eric A. Weiss, Niru Maheswaranathan, and Surya Ganguli. (2015). Deep Unsupervised Learning using Nonequilibrium Thermodynamics. arXiv preprint arXiv:1503.03585.